THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 1

Chapter 1: **Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $(-5, -2)$	Reflection in the y-axis changes the sign of the abscissa.
2	**(a)** 5	Apply the distance formula.
3	**(b)** $(2, 2)$	Use $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.
4	**(d)** $(6, 0)$	Ordinate is 0 on the x-axis.
5	**(d)** Quadrant II	Use the sign rules for the quadrants.
6	**(d)** 7	The distance from the x-axis is $\lvert y \rvert$.
7	**(a)** $(8, 11)$	$B = 2M - A$ coordinatewise.
8	**(a)** $(3, 8)$	Reflection in the y-axis changes the sign of the abscissa.
9	**(b)** 13	Apply the distance formula.
10	**(b)** $(-2, 4)$	Use $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$PQ = \sqrt{(3)^2 + (4)^2}$ (1); $PQ = 5$ units (1).	1+1
12	$\left(\frac{-3+(9)}{2}, \frac{7+(1)}{2}\right)$ (1); Midpoint $= (3, 4)$ (1).	1+1
13	$B = (2 \cdot 4 - (1),\ 2 \cdot 6 - (2))$ (1); $B = (7, 10)$ (1).	1+1
14	$PQ = \sqrt{(8)^2 + (-15)^2}$ (1); $PQ = 17$ units (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB^2 = 25$ (1); $BC^2 = 25$, $CA^2 = 64$ (1); Two sides are equal, so the triangle is isosceles (1).	1+1+1
16	Let the point be $(x, 0)$; set $PA^2 = PB^2$ (1); $(x - (-1))^2 + 16 = (x - 5)^2 + 16$ (1); $x = 2$, point $(2, 0)$ (1).	1+1+1
17	$D = (1, 29)$ (1); $AC = \sqrt{21^2 + 28^2}$ (1); $AC = 35$ units (1).	1+1+1
	OR: $AB^2 = 25$ (1); $BC^2 = 25$, $CA^2 = 64$ (1); Two sides are equal, so the triangle is isosceles (1).	1+1+1
18	Let the point be $(x, 0)$; set $PA^2 = PB^2$ (1); $(x - (-5))^2 + 4 = (x - 3)^2 + 4$ (1); $x = -1$, point $(-1, 0)$ (1).	1+1+1
	OR: $D = (-2, 35)$ (1); $AC = \sqrt{24^2 + 32^2}$ (1); $AC = 40$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AB^2 = 17$, $DC^2 = 17$: $AB = DC$ (1); $BC^2 = 13$, $AD^2 = 13$: $BC = AD$ (1); Midpoint of $AC = (3, 2)$ (1); Midpoint of $BD = (3, 2)$ (1); Opposite sides equal and diagonals bisect each other: parallelogram (1).	1+1+1+1+1
	OR: $AB = \sqrt{225} = 15$ (1); $BC = \sqrt{144} = 12$ (1); $CA = \sqrt{81} = 9$ (1); Perimeter $= 36$ units (2).	1+1+1+2
20	$AB^2 = 17$, $DC^2 = 17$: $AB = DC$ (1); $BC^2 = 13$, $AD^2 = 13$: $BC = AD$ (1); Midpoint of $AC = (4, 3)$ (1); Midpoint of $BD = (4, 3)$ (1); Opposite sides equal and diagonals bisect each other: parallelogram (1).	1+1+1+1+1
	OR: $AB = \sqrt{100} = 10$ (1); $BC = \sqrt{144} = 12$ (1); $CA = \sqrt{100} = 10$ (1); Perimeter $= 32$ units (2).	1+1+1+2

Mistakes to avoid:

- Writing the point as (y, x): the abscissa x always comes first, so $(2, 3) \neq (3, 2)$.
- Mixing up the signs of the quadrants, e.g. placing $(-3, 4)$ in Quadrant IV; Quadrant II is $(-, +)$.
- Reflecting in the x-axis but changing the sign of the abscissa (that is a reflection in the y-axis).
- Saying that $(0, -6)$ or $(5, 0)$ lies in a quadrant; points on the axes belong to no quadrant.
- In the distance formula, forgetting the square root or adding the coordinates instead of subtracting them.
- In the midpoint formula, dividing only one coordinate by 2, or subtracting instead of adding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $(-4, 0)$	Ordinate is 0 on the x-axis.
2	**(b)** Quadrant IV	Use the sign rules for the quadrants.
3	**(a)** 11	The distance from the x-axis is $\lvert y \rvert$.
4	**(c)** $(7, 0)$	$B = 2M - A$ coordinatewise.
5	**(d)** $(-7, 4)$	Reflection in the y-axis changes the sign of the abscissa.
6	**(c)** 17	Apply the distance formula.
7	**(c)** $(1, 5)$	Use $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.
8	**(a)** $(9, 0)$	Ordinate is 0 on the x-axis.
9	**(d)** Quadrant III	Use the sign rules for the quadrants.
10	**(b)** 2	The distance from the x-axis is $\lvert y \rvert$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\left(\frac{4+(-2)}{2}, \frac{-8+(2)}{2}\right)$ (1); Midpoint $= (1, -3)$ (1).	1+1
12	$B = (2 \cdot 0 - (-3),\ 2 \cdot 1 - (5))$ (1); $B = (3, -3)$ (1).	1+1
13	$PQ = \sqrt{(7)^2 + (24)^2}$ (1); $PQ = 25$ units (1).	1+1
14	$\left(\frac{-7+(3)}{2}, \frac{-5+(11)}{2}\right)$ (1); Midpoint $= (-2, 3)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB^2 = 25$ (1); $BC^2 = 25$, $CA^2 = 36$ (1); Two sides are equal, so the triangle is isosceles (1).	1+1+1
16	Let the point be $(x, 0)$; set $PA^2 = PB^2$ (1); $(x - (-2))^2 + 1 = (x - 8)^2 + 1$ (1); $x = 3$, point $(3, 0)$ (1).	1+1+1
17	$D = (0, 56)$ (1); $AC = \sqrt{11^2 + 60^2}$ (1); $AC = 61$ units (1).	1+1+1
	OR: $AB^2 = 25$ (1); $BC^2 = 25$, $CA^2 = 36$ (1); Two sides are equal, so the triangle is isosceles (1).	1+1+1
18	Let the point be $(x, 0)$; set $PA^2 = PB^2$ (1); $(x - (-2))^2 + 36 = (x - 2)^2 + 36$ (1); $x = 0$, point $(0, 0)$ (1).	1+1+1
	OR: $D = (3, 50)$ (1); $AC = \sqrt{14^2 + 48^2}$ (1); $AC = 50$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AB^2 = 17$, $DC^2 = 17$: $AB = DC$ (1); $BC^2 = 18$, $AD^2 = 18$: $BC = AD$ (1); Midpoint of $AC = (2.5, 1)$ (1); Midpoint of $BD = (2.5, 1)$ (1); Opposite sides equal and diagonals bisect each other: parallelogram (1).	1+1+1+1+1
	OR: $AB = \sqrt{100} = 10$ (1); $BC = \sqrt{64} = 8$ (1); $CA = \sqrt{36} = 6$ (1); Perimeter $= 24$ units (2).	1+1+1+2
20	$AB^2 = 10$, $DC^2 = 10$: $AB = DC$ (1); $BC^2 = 13$, $AD^2 = 13$: $BC = AD$ (1); Midpoint of $AC = (2.5, 2)$ (1); Midpoint of $BD = (2.5, 2)$ (1); Opposite sides equal and diagonals bisect each other: parallelogram (1).	1+1+1+1+1
	OR: $AB = \sqrt{169} = 13$ (1); $BC = \sqrt{25} = 5$ (1); $CA = \sqrt{144} = 12$ (1); Perimeter $= 30$ units (2).	1+1+1+2

Mistakes to avoid:

- Writing the point as (y, x): the abscissa x always comes first, so $(2, 3) \neq (3, 2)$.
- Mixing up the signs of the quadrants, e.g. placing $(-3, 4)$ in Quadrant IV; Quadrant II is $(-, +)$.
- Reflecting in the x-axis but changing the sign of the abscissa (that is a reflection in the y-axis).
- Saying that $(0, -6)$ or $(5, 0)$ lies in a quadrant; points on the axes belong to no quadrant.
- In the distance formula, forgetting the square root or adding the coordinates instead of subtracting them.
- In the midpoint formula, dividing only one coordinate by 2, or subtracting instead of adding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $(-4, 4)$	$B = 2M - A$ coordinatewise.
2	**(c)** $(9, -6)$	Reflection in the y-axis changes the sign of the abscissa.
3	**(d)** 25	Apply the distance formula.
4	**(d)** $(-5, -1)$	Use $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.
5	**(b)** $(-7, 0)$	Ordinate is 0 on the x-axis.
6	**(d)** negative y-axis	Use the sign rules for the quadrants.
7	**(b)** 13	The distance from the x-axis is $\lvert y \rvert$.
8	**(a)** $(-8, 2)$	$B = 2M - A$ coordinatewise.
9	**(b)** 10	Apply the distance formula.
10	**(c)** $(2, 4)$	Use $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$B = (2 \cdot 2 - (7),\ 2 \cdot 3 - (-4))$ (1); $B = (-3, 10)$ (1).	1+1
12	$PQ = \sqrt{(12)^2 + (5)^2}$ (1); $PQ = 13$ units (1).	1+1
13	$\left(\frac{6+(-10)}{2}, \frac{0+(8)}{2}\right)$ (1); Midpoint $= (-2, 4)$ (1).	1+1
14	$B = (2 \cdot -1 - (-6),\ 2 \cdot 4 - (-2))$ (1); $B = (4, 10)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB^2 = 25$ (1); $BC^2 = 25$, $CA^2 = 36$ (1); Two sides are equal, so the triangle is isosceles (1).	1+1+1
16	Let the point be $(x, 0)$; set $PA^2 = PB^2$ (1); $(x - (-7))^2 + 25 = (x + 1)^2 + 25$ (1); $x = -4$, point $(-4, 0)$ (1).	1+1+1
17	$D = (-5, 35)$ (1); $AC = \sqrt{27^2 + 36^2}$ (1); $AC = 45$ units (1).	1+1+1
	OR: $AB^2 = 25$ (1); $BC^2 = 25$, $CA^2 = 36$ (1); Two sides are equal, so the triangle is isosceles (1).	1+1+1
18	Let the point be $(x, 0)$; set $PA^2 = PB^2$ (1); $(x - (4))^2 + 9 = (x - 6)^2 + 9$ (1); $x = 5$, point $(5, 0)$ (1).	1+1+1
	OR: $D = (2, 40)$ (1); $AC = \sqrt{30^2 + 40^2}$ (1); $AC = 50$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AB^2 = 17$, $DC^2 = 17$: $AB = DC$ (1); $BC^2 = 13$, $AD^2 = 13$: $BC = AD$ (1); Midpoint of $AC = (1, 0)$ (1); Midpoint of $BD = (1, 0)$ (1); Opposite sides equal and diagonals bisect each other: parallelogram (1).	1+1+1+1+1
	OR: $AB = \sqrt{169} = 13$ (1); $BC = \sqrt{576} = 24$ (1); $CA = \sqrt{169} = 13$ (1); Perimeter $= 50$ units (2).	1+1+1+2
20	$AB^2 = 26$, $DC^2 = 26$: $AB = DC$ (1); $BC^2 = 13$, $AD^2 = 13$: $BC = AD$ (1); Midpoint of $AC = (4.5, 0)$ (1); Midpoint of $BD = (4.5, 0)$ (1); Opposite sides equal and diagonals bisect each other: parallelogram (1).	1+1+1+1+1
	OR: $AB = \sqrt{25} = 5$ (1); $BC = \sqrt{64} = 8$ (1); $CA = \sqrt{25} = 5$ (1); Perimeter $= 18$ units (2).	1+1+1+2

Mistakes to avoid:

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- Mixing up the signs of the quadrants, e.g. placing $(-3, 4)$ in Quadrant IV; Quadrant II is $(-, +)$.
- Reflecting in the x-axis but changing the sign of the abscissa (that is a reflection in the y-axis).
- Saying that $(0, -6)$ or $(5, 0)$ lies in a quadrant; points on the axes belong to no quadrant.
- In the distance formula, forgetting the square root or adding the coordinates instead of subtracting them.
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